

17.  $\Sigma$  为对称矩阵, 可作正交变换对角化

$\therefore \exists A$  阵, 使得  $A \Sigma A^T = D$   $D$  为对角矩阵,  $= \text{diag}\{d_1, d_2, \dots, d_n\}$

$\therefore$  故知,  $Y_i \sim N(0, D)$ ,  $\text{Cov}(Y_i, Y_j) = 0$  ( $i \neq j$ )

$\therefore Y_i$  互相独立, 且  $Y_i \sim N(0, d_i)$

18.

$$(11) \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & \frac{5}{5} \\ \frac{4}{5} & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} \frac{113}{5} & \frac{112}{5} \\ \frac{112}{5} & \frac{113}{5} \end{pmatrix}$$

12)  $\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 7 \\ 8 \end{pmatrix}, \begin{pmatrix} \frac{112}{5} & \frac{112}{5} \\ \frac{112}{5} & \frac{113}{5} \end{pmatrix}\right)$

$Y_2 | Y_1$  也为正态分布, 其期望为:  $8 + \frac{112}{5} \times \frac{5}{113} \times (Y_1 - 7) = 8 + \frac{112}{113}(Y_1 - 7)$

$Y_1 + Y_2 = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$   $\therefore E(Y_1 + Y_2) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \end{pmatrix} = 15$

13)  ~~$E(X_2 | X_1)$~~   $X_2 | X_1 \sim N\left(2 + \frac{4}{5} X_1 (X_1 - 1), 1 - \left(\frac{4}{5}\right)^2 X_1\right)$

$\therefore E(X_2 | X_1) = 2 + \frac{4}{5}(X_1 - 1)$   $D(X_2 | X_1) = \frac{9}{25}$

(4, 显然二者为高斯分布, 故只需证明二者协方差为 0

$E(X_2 - E(X_2 | X_1)) = E X_2 - E(E(X_2 | X_1)) = E X_2 - E X_2 = 0$

$E(X_1 X_2 - X_1 E(X_2 | X_1)) = E X_1 X_2 - E(X_1 (2 + \frac{4}{5}(X_1 - 1))) = \text{Cov}(X_1, X_2) + E X_1 E X_2 - E(2 X_1 + \frac{4}{5} X_1^2 - \frac{4}{5} X_1)$   
 $= 0$

$\therefore \text{Cov}(X_1, X_2 - E(X_2 | X_1)) = 0$   $\therefore$  独立.

20. (a) 必要性  $(X_{t_1}, X_{t_2}, \dots, X_{t_n})^T$  服从高斯分布

~~Cov~~ 且与  $(X_{t_1+a}, X_{t_2+a}, \dots, X_{t_n+a})^T$  相同

$\therefore E X_{t_1+a} = E X_{t_1}$  由  $t_1$  的任意性  $\therefore E X_i$  为常数

$$\text{Cov}(X_{t_1}, X_{t_2}) = \text{Cov}(X_{t_1+a}, X_{t_2+a})$$

由  $a$  的任意性  $\text{Cov}(X_{t_1}, X_{t_2})$  只与  $t_1, t_2$  有关.

充分性: 考虑  $(X_{t_1+a}, X_{t_2+a}, \dots, X_{t_n+a})^T$  联合分布的特征函数.

$$\varphi(\vec{\theta}) = \exp\left\{i \vec{\theta}^T \vec{\mu} - \frac{1}{2} \vec{\theta}^T \Sigma \vec{\theta}\right\}$$

而  $\vec{\mu}$  与  $a$  无关.  $\Sigma$  中元素为  $\text{Cov}(X_{t_i+a}, X_{t_j+a})$  只与  $|t_i - t_j|$  有关.

$\therefore \varphi(\vec{\theta})$  与  $a$  无关.

$$\therefore \text{对任意 } a, \varphi_{X_{t_1}, X_{t_2}, \dots, X_{t_n}}(\vec{\theta}) = \varphi_{X_{t_1+a}, X_{t_2+a}, \dots, X_{t_n+a}}(\vec{\theta})$$

$\therefore$  二者有相同的联合分布.

$$(b) E U_t = e^{-\frac{\sigma^2 t}{2}} E X_{e^{\sigma t}} = 0$$

$$\text{Cov}(X_s, X_t) = E X_s X_t$$

$$\begin{aligned} \text{Cov}(U_s, U_t) &= E U_s U_t = e^{-\frac{\sigma^2 s}{2} - \frac{\sigma^2 t}{2}} E X_{e^{\sigma s}} X_{e^{\sigma t}} \\ &= e^{-\frac{\sigma^2 s}{2} - \frac{\sigma^2 t}{2}} \text{Cov}(X_{e^{\sigma s}}, X_{e^{\sigma t}}) \\ &= e^{-\frac{\sigma^2 (s+t)}{2}} e^{\frac{\sigma^2 \min(s, t)}{2}} \\ &= e^{-\frac{\sigma^2}{2} |t-s|} \end{aligned}$$

$\therefore \{U_t, t \geq 0\}$  为平稳高斯过程

$$21. (1), \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 2 & 2 & \dots & 2 \\ \vdots & 2 & 3 & \dots & \vdots \\ 1 & 2 & \vdots & \dots & n \end{pmatrix}\right)$$

$$\beta_1 + \beta_2 + \dots + \beta_n = (1, 1, \dots, 1) \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} \sim N\left(0, \frac{n(n+1)(2n+1)}{6}\right)$$

$$(2) E(\beta_{t+1} - \beta_t) = 0$$

$$\text{Cov}(Y_s, Y_t) = E Y_s Y_t = E[(\beta_{s+1} - \beta_s)(\beta_{t+1} - \beta_t)]$$

$$= \min(s+1, t+1) + \min(s, t) - \min(s, t+1) - \min(t, s+1)$$

$$= \frac{1}{2}(|s-t-1| + |s-t+1| - 2|s-t|)$$

~~不是常数~~ 只依赖于  $|s-t|$

$\therefore$  是平稳过程

$$23. \begin{pmatrix} \beta_s \\ \beta_t \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} s & s \\ s & t \end{pmatrix}\right) \quad \therefore \beta_s | \beta_t \sim N\left(\frac{s}{t}(\beta_t - 0), \frac{s - \frac{s^2}{t}}{t}\right)$$

$$\therefore E(\beta_s | \beta_t = x) = \frac{xs}{t}$$

$$12. F(x) = P(M_n \leq x) = 1 - P(M_n > x)$$

当  $x < a$  时, 上式  $= 0$

$$\text{当 } x \geq a \text{ 时, 上式} = 1 - (1 - e^{-(x-a)})^n \rightarrow 1$$

$$\therefore M_n \xrightarrow{D} F_a \quad \therefore M_n \xrightarrow{P} a$$

解法二:

$$Y = \begin{pmatrix} Y \\ Z \end{pmatrix} = \begin{pmatrix} A \\ C \end{pmatrix} X + \begin{pmatrix} B \\ D \end{pmatrix}$$

$$\therefore \begin{pmatrix} Y \\ Z \end{pmatrix} \sim N \left( \begin{pmatrix} A \\ C \end{pmatrix} \mu + \begin{pmatrix} B \\ D \end{pmatrix}, \begin{pmatrix} A \Sigma A^T & A \Sigma C^T \\ C \Sigma A^T & C \Sigma C^T \end{pmatrix} \right)$$

$Y, Z$  相互独立  $\Leftrightarrow Y$  的分量与  $Z$  的分量独立

$$\Leftrightarrow C \Sigma A^T = 0$$

$$\therefore \text{由 } E X = 0 \quad X = (3 \quad 2 \quad -1) \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix} \sim N(0, 18)$$

$$\therefore E X = 0 \quad D X = 18 \quad f_X(x) = e^{-9x^2}$$

$$(2) \quad X, Y \text{ 相互独立} \Leftrightarrow \text{Cov}(X, Y) = 0 \Leftrightarrow E X Y = 0$$

$$E X Y = E (3B_1 + 2B_2 - B_3)(4B_1 + B_2) = 2C + 5 \approx 0$$

$$C = -\frac{5}{2}$$

$$(3) \quad E(X | B_2 = 1) = 3E(B_1 | B_2 = 1) + 2 - E(B_3 | B_2 = 1)$$

$$= \frac{5}{2}$$

$$\text{三、 } EX^2 = 1 \quad EY^2 = 1$$

$$EX^4 = 3 \quad EY^4 = 3$$

$$\therefore DX^2 = DY^2 = 2$$

$$EX^2Y^2 = E(E(X^2Y^2|Y))$$

$$= E(Y^2 E(X^2|Y))$$

$$\text{而 } X|Y \sim N(\rho Y, 1-\rho^2) \quad \therefore E(X^2|Y) = (1-\rho^2) + \rho^2 Y^2$$

$$\therefore \text{上式} = \cancel{E(Y^2)} (1-\rho^2) EY^2 + \rho^2 EY^4$$

$$= 1 + 2\rho^2$$

$$\therefore \rho_{X^2, Y^2} = \frac{EX^2Y^2 - EX^2EY^2}{\sqrt{DX^2DY^2}} = \rho^2$$